DERIVED HOM FROM A K-PROJECTIVE DIFFERENTIAL GRADED MODULE IS DG-HOM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: dependencies carried as navigation stubs: Lemma lemma-derived-restriction (construction), Example example-dgm-dg-cat (degree-wise Hom_K identification), Lemma lemma-hom-derived (property (P) case); all in-source.

Editorial note. The label lemma-compute-derived-restriction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Construction of derived restriction. Same unit, Lemma lemma-derived-restriction. By construction, RHom(N, M) is computed by choosing a quasi-isomorphism M -> I with I a differential graded B-module of property (I) and setting RHom(N, M) = Hom in Mod^dg_(B,d) from N to I.
- Hom-complex cohomology identification. Same unit, Example example-dgm-dg-cat and Lemma lemma-hom-derived. The cohomology in each degree of the Hom-complex between differential graded modules is Hom in the homotopy category after shifting (standard dg-category fact recorded in the example); the property (P) case of the hypothesis is supplied by Lemma lemma-hom-derived.

Source reference. Original reference 09LC: sources/stacks-dga/pinned-original-source.tex, source lines 3525–3525. Prerequisite pointer invoked at proof line(s) 5407; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09LI: sources/stacks-dga/pinned-original-source.tex, source lines 5272–5272. Prerequisite pointer invoked at proof line(s) 5396; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KY: sources/stacks-dga/pinned-original-source.tex, source lines 2850–2850. Prerequisite pointer invoked at proof line(s) 5409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let N be a differential graded (A, B)-bimodule. If $\mathrm{Hom}_{D(B,d)}(N, N') = \mathrm{Hom}_{K(Mod_{(B,d)})}(N, N')$ for all $N' \in K(B, d)$, for example if N has property (P) as a differential graded B-module, then*

$$R\,\mathrm{Hom}(N, M) = \mathrm{Hom}_{Mod^{dg}_{(B,d)}}(N, M)$$

functorially in M in $D(B, d)$.

Proof. By construction (Lemma 09LI) to find $R\,\mathrm{Hom}(N, M)$ we choose a quasi-isomorphism $M \to I$ where I is a differential graded B-module with property (I) and we set $R\,\mathrm{Hom}(N, M) = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, I)$. By assumption the map

$$\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, M) \longrightarrow \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, I)$$

induced by $M \to I$ is a quasi-isomorphism, see discussion in Example 09LC. This proves the lemma. If N has property (P) as a B-module, then we see that the assumption is satisfied by Lemma 09KY. □